

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			blood passes through the heart twice (1) during each full circuit/ cycle (1) Linked to 1 st mark point OR blood must pass through pulmonary (1) and systemic system (1)	2			2		
	(b)	(i)		Bicuspid (valve)/left atrio-ventricular valve/ mitral/	1			1		1
	(c)			Any three (x1) from: Valve opens when heart/ ventricle contracts (1) Valve closes when heart/ ventricle relaxes (1) Prevents backflow (of blood) (1) (from aorta) to (left) ventricle (1)		3		3		
	(d)			left ventricle pumps blood to the body(1) right ventricle pumps blood to the lungs/ not as far}(1)	2			2		
	(e)			heart is a muscle (1) (exercise) increases heart size/ heart gets stronger with (exercise)/ (exercise) builds up heart size (1) Heart becomes more muscular = 2 marks			2	2		
	(f)	(i)		CO 5 minutes = $70 \times 70 = 4900$ + CO 20 minutes = $110 \times 70 = 7700$ (1) (1 mark for calculating both cardiac outputs)		1		1	1	
		(ii)		$57/57.1/57.142857 = 2$ marks If incorrect award 1 mark for $2800/4900 \times 100$ Incorrect rounding of above answer ECF from (i)		2		2	2	
				Question 4 total	5	6	2	13	3	1